

Predicting problematic internet use from tabular health and activity data

Anonymous technical report. All numeric results and plots below are generated from the current experiment artifacts, not entered by hand.

1. Problem statement

The task is to predict the ordinal Severity Impairment Index (sii) from the tabular Child Mind Institute - Problematic Internet Use competition data [1]. The features describe demographics, physical measurements, physical activity, fitness, body composition, sleep, and reported computer/internet use. The label has four ordered classes: 0 (none), 1 (mild), 2 (moderate), and 3 (severe). This is a predictive task in the internet-use/physical-activity domain, not an estimate of a causal effect of internet use on physical activity.

The raw training table contains 3960 participants: 2736 labeled and 1224 unlabeled. Class counts are 0: 1594, 1: 730, 2: 378, 3: 34. The example test contains 20 rows.

Only labeled participants are used for supervised fitting and validation. Missing labels are not assigned a class or pseudo-labeled. PCIAT questionnaire items, total score, and season are excluded because the label is derived from the questionnaire total. Participant ID is only used for alignment and stable fold assignment. The public example test file has no usable evaluation labels; producing a submission is not evidence of hidden-test or leaderboard accuracy.

The primary metric is quadratic weighted kappa (QWK), using the fixed class scale 0, 1, 2, 3. With true classes y and predicted classes c , it is $1 - \text{observed} / \text{expected}$, where $\text{observed} = \sum((y-c)^2)$, and $\text{expected} = \sum(y^2) + \sum(c^2) - 2 * \text{mean}(y) * \sum(c)$. Larger ordinal errors receive larger penalties. Class precision, recall, F1, prediction counts, and confusion matrices supplement QWK; aggregate agreement alone can hide rare-class errors.

2. Baselines

We test the standard implementations provided by scikit-learn, CatBoost, and XGBoost [2-5]; these libraries and algorithms are not claimed as our own inventions. The experiment includes a majority-class predictor, Ridge with fixed midpoint rounding, and single CatBoost, XGBoost, ExtraTrees, and Ridge regressors with training-only QWK calibration. Each untuned member uses candidate 0 from the explicit grid in `src/experiment.py`.

The reference equal ensemble averages the four untuned continuous predictions before calibration. An additional strong reference chooses a tuned single member using inner-OOF QWK, never the outer-fold score. This avoids choosing the best reference retrospectively on validation labels.

All current comparisons use the same participants, feature preparation, outer folds, and QWK definition. Historical notebook 07/08/early-09 winning scores are exploratory and are not used to choose current numerical hyperparameters or assert a current improvement. The earlier global tuning and calibration protocol differs from the current nested procedure.

3. Proposed model and training procedure

The proposed architecture is an equal-weight ensemble of four regressors: CatBoost (categorical boosting [3]), XGBoost (histogram tree boosting [4]), ExtraTrees (randomized trees [5]), and Ridge (regularized linear regression [2]). Its continuous score is 0.25 times the sum of member scores. The three strictly ordered cut points map that score to classes 0-3. Equal averaging provides different inductive biases without adding learned blend weights to the primary architecture. The contribution is this trained ordinal ensemble and its evaluation/calibration pipeline, not a novel boosting algorithm.

Layer A preprocessing is deterministic and row-local. It checks enumerated codes against the competition dictionary, applies fixed heuristic plausibility bounds, recomputes BMI using the dictionary's inch/pound units, merges child and adolescent PAQ totals, and flags blocks with all measurement results missing. Season metadata does not count as a measurement. These flags do not infer whether a questionnaire was actually administered. The plausibility bounds and BMI discrepancy tolerances are project heuristics, not clinically validated diagnostic rules; they must not be used to make clinical decisions.

CatBoost retains numeric NaNs and categorical strings; categorical missing values become an explicit level. Other members use training-partition median imputation and one-hot encoding; Ridge additionally uses standard scaling. The XGBoost member also receives five deterministic interactions: treadmill duration, waist/height ratio, pulse pressure, a raw fitness composite, and BMI/age. The fitness composite combines heterogeneous measurements and is not a clinical score. An entirely missing fitness block remains NaN, not zero.

The measured run uses 2 repeated ID-stable stratified outer CV runs, with seeds [42, 2026], 5 outer folds, and 3 inner folds per outer training partition. CPU pools are limited to 4 threads. Early stopping has patience 60.

Within each outer training partition, each member's generic candidate grid is evaluated on three inner stratified folds. Candidate capacity/regularization is chosen by inner-OOF RMSE. QWK cut points are then calibrated on the selected inner-OOF scores. The coordinate search evaluates all distinct score cut points for each boundary and uses multiple starting points; it is a local optimizer, not a guarantee of a global optimum. Hyperparameter selection and calibration may overfit the inner data; only the untouched outer score evaluates their combined effect [6]. No previously winning Optuna parameters are reused.

For boosting, each individual training partition is split 85/15 for early stopping. XGBoost preprocessing is learned on the 85% only. A fresh model and fresh preprocessing are refit on the entire available partition with the selected tree count. Outer validation labels never decide the tree count, medians, vocabularies, scaling, capacity, weights, or class thresholds.

For deployment, the same inner-selection/calibration algorithm is run on all labeled training participants, followed by final fits on all labeled rows. Weights remain exactly 0.25 for each member. The saved model bundle reproduces the submission after serialization and reload. The primary policy is specified in code, not switched to a retrospectively higher full-OOF blend score.

4. Model improvements and controlled comparisons

The improvement over the untuned proposed ensemble is inner-fold selection of tree capacity, regularization, sampling, Ridge alpha, and training-fold percentile clipping for Ridge. Clipping bounds are learned on training data only; they protect a scale-sensitive linear member against extreme input values. Both ensemble versions receive the same row-local features and inner-only rounding procedure; the comparison isolates the adaptive member hyperparameter selection procedure. The search has three candidates per member and a fixed, reproducible budget. The full grids, selected candidates, inner RMSE, and stopping tree counts are saved in results/nested_metadata.json and results/nested_tuning_results.csv.

A secondary weighted ensemble searches nonnegative weights in steps of 0.25 and calibrates them on inner OOF only. Its outer scores are reported but it is not the primary exported policy. We do not assume that the most complex ensemble or hyperparameter tuning must improve QWK.

reference	delta_qwk	ci_low	ci_high
Inner-selected single model	0.0091	-0.0031	0.0213
Baseline equal ensemble	0.0107	-0.0009	0.0227
Tuned xgboost	0.0068	-0.0061	0.0202

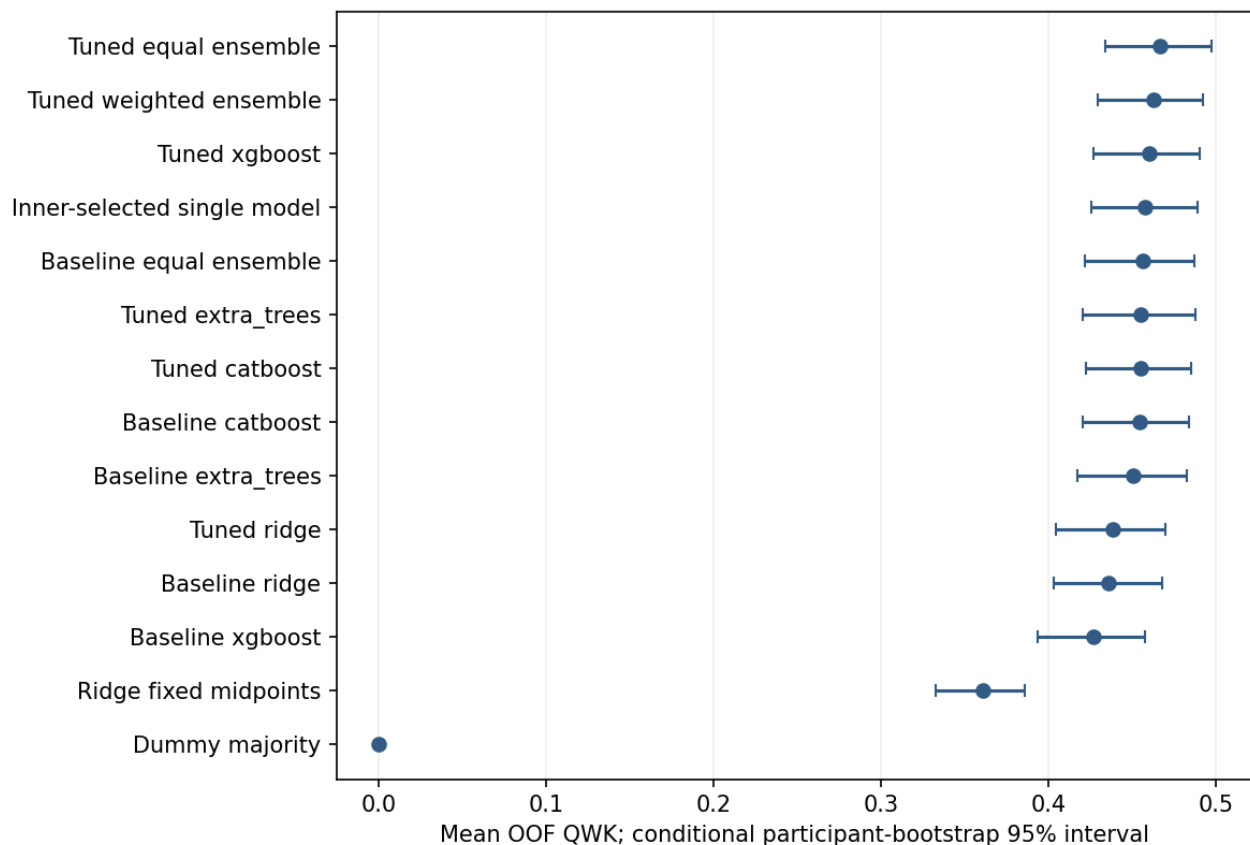
Against Inner-selected single model, the observed change is +0.0091; its interval includes zero; a reliable gain is not established. Against Baseline equal ensemble, the observed change is +0.0107; its interval includes

zero; a reliable gain is not established. Against Tuned xgboost, the observed change is +0.0068; its interval includes zero; a reliable gain is not established.

5. Results and error analysis

The following table reports the mean whole-OOF QWK across repeated nested CV runs, not an average obtained by concatenating repeat rows as independent participants. Fold-level and repeat-level results are saved separately.

setup	mean_oof_qwk	ci_low	ci_high
Tuned equal ensemble	0.4669	0.4340	0.4973
Tuned weighted ensemble	0.4627	0.4293	0.4926
Tuned xgboost	0.4601	0.4270	0.4904
Inner-selected single model	0.4579	0.4255	0.4890
Baseline equal ensemble	0.4562	0.4219	0.4873
Tuned extra_trees	0.4555	0.4208	0.4880
Tuned catboost	0.4553	0.4226	0.4852
Baseline catboost	0.4544	0.4208	0.4840
Baseline extra_trees	0.4504	0.4173	0.4825
Tuned ridge	0.4382	0.4045	0.4698
Baseline ridge	0.4362	0.4031	0.4679
Baseline xgboost	0.4268	0.3934	0.4578
Ridge fixed midpoints	0.3609	0.3330	0.3857
Dummy majority	0.0000	0.0000	0.0000

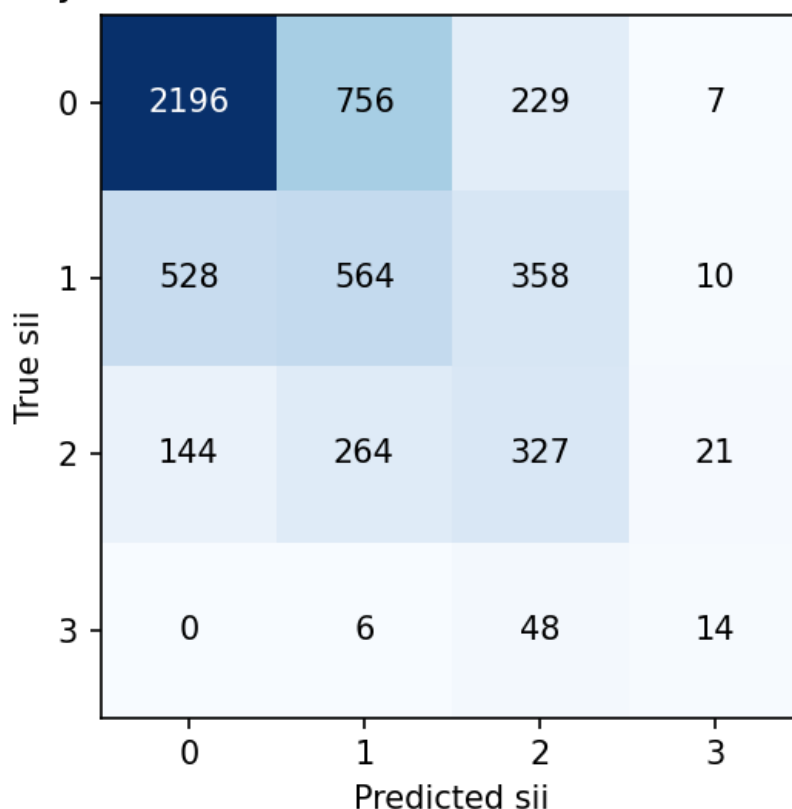


The intervals are percentile bootstrap diagnostics conditional on the fixed OOF predictions. The same sampled participant indices are used in every repeat and in both sides of every comparison. This accounts for participant pairing and avoids treating repeated predictions as independent observations. It does not repeat model fitting or all historical design decisions; it is not a full retraining or post-selection confidence interval. Two random seeds provide a limited sensitivity check, not exhaustive uncertainty quantification.

class	precision	recall	f1-score	true_positives
0	0.7657	0.6888	0.7252	1098.0
1	0.3548	0.3863	0.3698	282.0
2	0.3400	0.4325	0.3807	163.5
3	0.2692	0.2059	0.2333	7.0

Across repeats, severe-class recall is 0.206, with a mean 7.0 true positives out of 34 severe participants per repeat. The table averages class metrics across repeats; fractional true-positive counts are repeat means, not fractional participants. Confusion-matrix counts sum across repeats and are not counts of unique children.

Primary ensemble: counts summed across CV repeats



Errors and rare-class performance are reported even when they weaken the headline result. The model is not presented as ready for diagnosis or clinical screening. The small severe-class sample limits conclusions and motivates more labeled examples and external validation rather than unsupported accuracy claims. Excluding unlabeled participants can induce selection bias because their feature missingness differs; the evaluated population is the labeled cohort, not every child or every competition participant.

6. Reproducibility

Reference runtime: python 3.13.5, numpy 2.1.3, pandas 2.3.3, scikit-learn 1.6.1, scipy 1.15.3, catboost 1.2.10, xgboost 3.4.1, optuna 5.0.0, pyarrow 24.0.0, joblib 1.6.0, threadpoolctl 3.7.0.

The reference environment was installed in a new virtual environment, checked with pip check, and tested before training. requirements.txt pins direct dependencies; requirements.lock.txt also pins all resolved transitive packages for CPython 3.13.5 on Linux x86_64. Other Python/platform combinations are not claimed to reproduce the exact numerical run. Source and raw-data SHA256 digests, grids, seeds, fold selections, runtime versions, and result artifact digests are recorded in the metadata. Notebook cache is accepted only when the configuration, data, source, versions, and aggregate artifacts match.

Reproduce from the extracted anonymous source directory:

```
python3.13 -m venv .venv
.venv/bin/python -m pip install -r requirements.lock.txt
.venv/bin/python -m pip check
.venv/bin/python -m unittest discover -s tests -v
.venv/bin/python -m src.download_data
.venv/bin/python -m src.experiment
.venv/bin/python -m src.verify
.venv/bin/python -m src.report
.venv/bin/python -m src.package
```

Before downloading, the reviewer must accept the competition rules and provide their own Kaggle credentials [1], as detailed in README. Access-controlled CSVs, participant-level OOF records, and model binaries are excluded from the source archive; the download/training commands recreate them. No private pretrained weights are required. The experiment builds processed features from raw CSVs and does not rely on a stale local parquet. The exported notebook is optional; the command-line workflow requires no manually reconstructed notebook state.

The technical report and plots are generated from verified aggregate artifacts. The anonymous archive excludes Git history, local environments, credentials, personal paths, and notebook outputs/metadata. Packaging uses an explicit allowlist and an automated path/identity-pattern scan. A final human anonymity inspection is still recommended; an automated scan is not proof that every possible identifying clue has been excluded.

7. Conclusion and limitations

The prespecified tuned equal ensemble achieves mean nested OOF QWK 0.4669, with a conditional bootstrap interval [0.4340, 0.4973]. Against Inner-selected single model, the observed change is +0.0091; its interval includes zero; a reliable gain is not established. Against Baseline equal ensemble, the observed change is +0.0107; its interval includes zero; a reliable gain is not established. Against Tuned xgboost, the observed change is +0.0068; its interval includes zero; a reliable gain is not established. The project implements several baselines, a trained proposed ensemble, and explicit inner-only hyperparameter modifications with measured outcomes; these are not claims of universal superiority.

The current procedure nests algorithmic hyperparameter selection and calibration inside outer training folds. However, project design followed exploratory analysis of this same dataset. We therefore do not describe these resampling estimates as a completely new untouched external cohort. Future work should use additional labeled data or an independently held-out cohort, especially for severe cases, and assess transportability across age and sex. Actigraphy is not used: it is a separate modality with incomplete coverage and was excluded to keep this tabular experiment reproducible on CPU. This scope choice may limit attainable performance and is not claimed to be optimal.

References and attribution

- [1] Child Mind Institute - Problematic Internet Use. Competition data, evaluation task, rules, and data dictionary. <https://www.kaggle.com/competitions/child-mind-institute-problematic-internet-use>
- [2] scikit-learn documentation. Ridge regression, preprocessing, stratified cross-validation, and metric implementations. <https://scikit-learn.org/1.6/>
- [3] Dorogush, Ershov, and Gulin. CatBoost: gradient boosting with categorical features support. <https://arxiv.org/abs/1810.11363>
- [4] Chen and Guestrin. XGBoost: A Scalable Tree Boosting System. <https://arxiv.org/abs/1603.02754>
- [5] scikit-learn. ExtraTreesRegressor implementation and references. <https://scikit-learn.org/1.6/modules/generated/sklearn.ensemble.ExtraTreesRegressor.html>
- [6] scikit-learn. Nested versus non-nested cross-validation. https://scikit-learn.org/1.6/auto_examples/model_selection/plot_nested_cross_validation_iris.html

All baseline/member algorithms are used through the attributed packages, not copied and claimed as new implementations. Figures are generated from this project's computed results. The composite architecture, glue code, tests, and report were developed with AI assistance; all numeric claims are checked against executed artifacts. This disclosure is not a claim of algorithmic novelty.